



EAST WEST UNIVERSITY

Lab Report-4

Submitted By:

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Section: 01

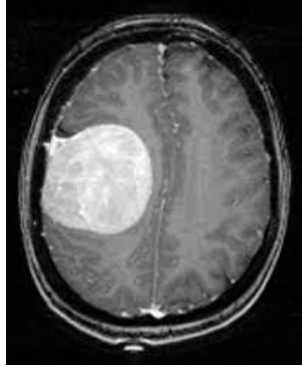
Submitted To:

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1. Sharpen the following image by applying the following and find out which one is better:

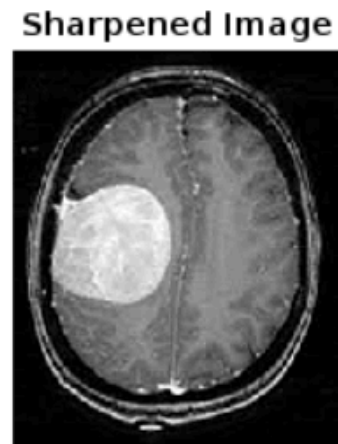
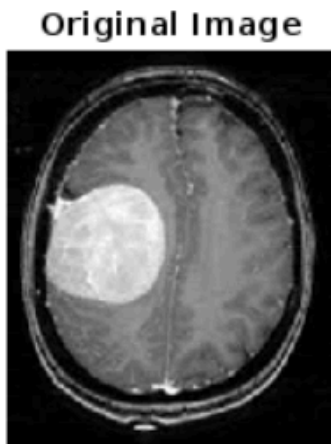
- a) Unsharp Masking
- b) High Boost Filtering



Answer:

```
a) img = imread('lab_4_1.jpg');  
j=imsharpen(img);  
subplot(2,2,1), imshow(img), title('Original Image')  
subplot(2,2,2), imshow(j), title('Sharpened Image');
```

Output:



b) Answer:

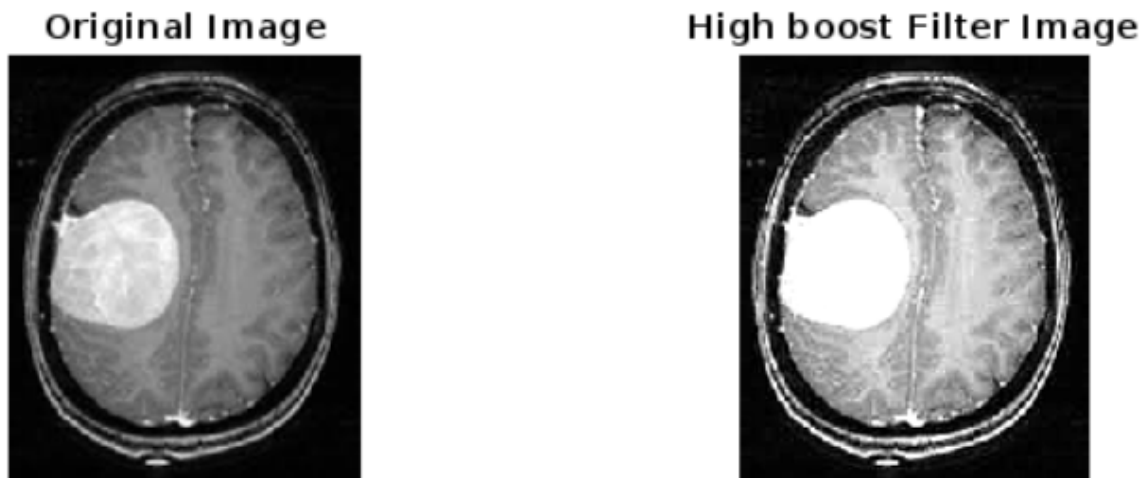
```
img = imread('lab_4_1.jpg');  
% Convert the input image to double datatype  
double_img = im2double(img);  
% Use the Laplacian filter to highlight edges  
Laplacian_filter= fspecial('laplacian', 0.5);
```

```

filtering = imfilter(double_img, Laplacian_filter, 'replicate');
filtered_img = double_img - filtering;
% Selecting a scaling factor for high boost filtering
scaling_factor=0.5;
% Applying high boost filtering
sharpen_img= double_img + scaling_factor* filtered_img;
subplot(2,2,1), imshow(img), title('Original Image');
subplot(2,2,2), imshow(sharpen_img), title('High boost Filter Image');

```

Output:



Unsharp mask is used to sharpen an image. The Unsharp Mask increases the image contrast along the edges of objects in an image. The high boost filter is used to enhance the high frequency components. We can sharpen edges of an image through the amplification and obtain a more clear image. It allows sharpening in high detail areas but little or no sharpening in flat or smooth areas.

Unsharp masking brings out more details by subtracting a blurry version of an image whereas high boost filtering allows sharpening areas but little or no sharpening in flat or smooth areas.

So, here Unsharp masking is better.

2. Sharpen the following image using the concept of Laplacian Filtering.



Answer:

```
img= imread('lab_4_2.jpg');
%apply laplacian filter
laplacianFilter= fspecial('laplacian');
laplacianImage= imfilter(img, laplacianFilter, 'replicate');
%subtract laplacian filter to sharpen the image
shar_img= img-laplacianImage;
subplot(2,3,1), imshow(img), title('Original Image');
subplot(2,3,2), imshow(laplacianImage, []), title('Laplacian Filtered Image');
subplot(2,3,3), imshow(shar_img), title('Sharpened Image');
```

Output:

Original Image



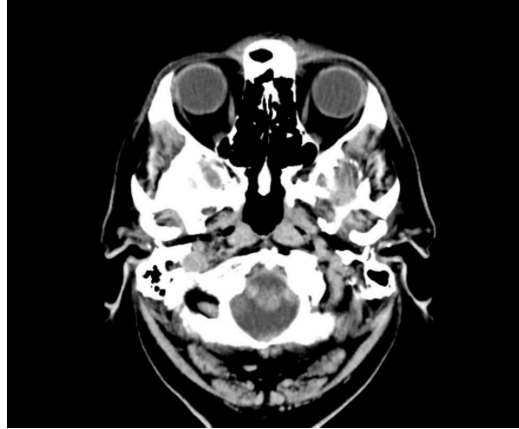
Laplacian Filtered Image



Sharpened Image



3. Use Roberts-cross, Sobel, and Prewitt operators to detect the edge of the following image.



Answer:

```
img= imread('lab_4_3.jpg');
imshow(img);
% Convert the image to grayscale
if size(img, 3) == 3
    grayImage = rgb2gray(img);
else
    grayImage = img;
end
% Apply Robert Cross operator for edge detection
robert_img = edge(grayImage, 'roberts');
% Apply sobel for edge detection
sobel_img = edge(grayImage, 'sobel');
% Apply prewitt for edge detection
pwitter_img = edge(grayImage, 'prewitt');
% Display the original and edge-detected images
subplot(2,4,1);
imshow(grayImage);
title('Original Image');
subplot(2,4,2);
imshow(robert_img);
title('Robert Cross Image');
subplot(2,4,3);
imshow(sobel_img);
title('Sobel Image');
subplot(2,4,4);
imshow(pwitter_img);
title('Prewitt Image');
```

Output:



4. Show performance comparison among High Boost, Unsharp, Laplacian Roberts-cross, Sobel, Prewitt and Canny filtering for edge detection – find out which one is better for the given image.

Answer:

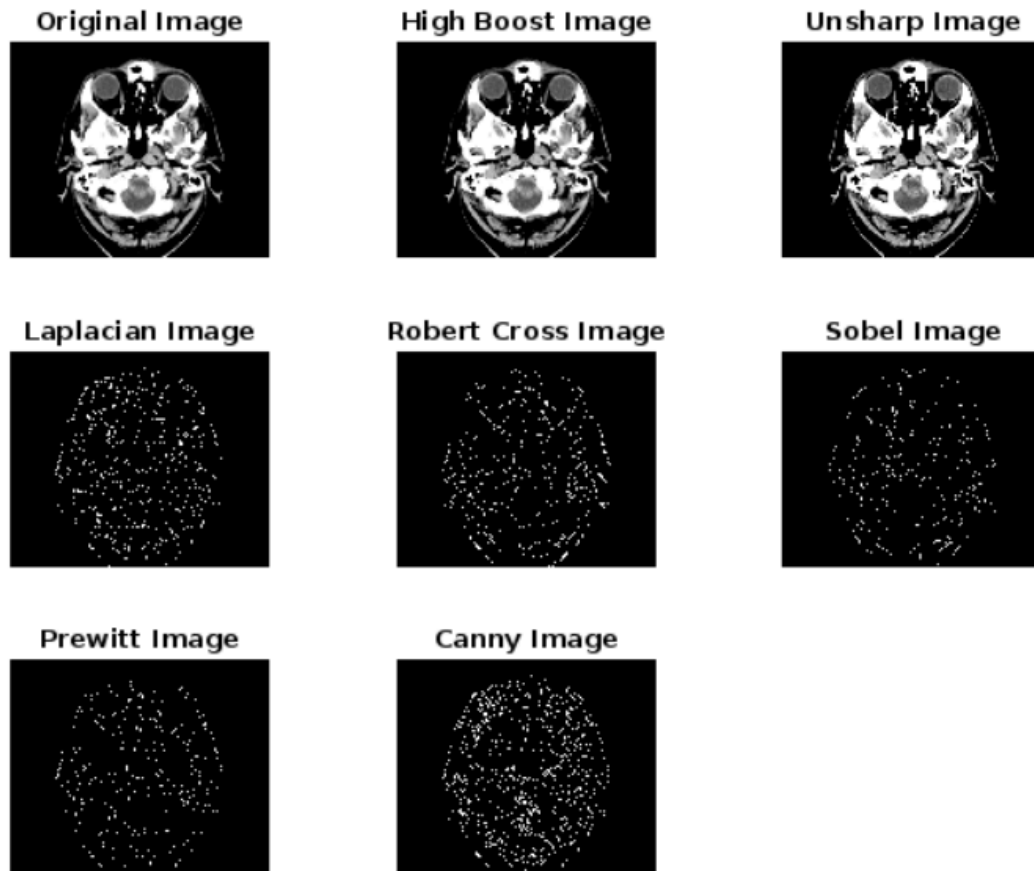
```
img= imread('lab_4_3.jpg');
imshow(img);
% Convert the image to grayscale
if size(img, 3) == 3
    grayImage = rgb2gray(img);
else
    grayImage = img;
end
% Apply high boost for edge detection
highboost_img = imsharpen(grayImage);
% Apply unsharp for edge detection
unsharp_img = imsharpen(grayImage, 'Radius', 2, 'Amount', 1.5);
% Apply laplacian for edge detection
laplacian_img= edge(grayImage, 'log');
% Apply Robert Cross operator for edge detection
robert_img = edge(grayImage, 'roberts');
% Apply sobel for edge detection
sobel_img = edge(grayImage, 'sobel');
% Apply prewitt for edge detection
pwitter_img = edge(grayImage, 'prewitt');
% Apply prewitt for edge detection
canny_img = edge(grayImage, 'canny');
% Display the original and edge-detected images
subplot(3,3,1);
imshow(grayImage);
title('Original Image');
subplot(3,3,2);
imshow(highboost_img);
title('High Boost Image');
subplot(3,3,3);
imshow(unsharp_img);
title('Unsharp Image');
subplot(3,3,4);
imshow(laplacian_img);
title('Laplacian Image');
subplot(3,3,5);
imshow(robert_img);
title('Robert Cross Image');
subplot(3,3,6);
imshow(sobel_img);
title('Sobel Image');
subplot(3,3,7);
```

```

imshow(pwitter_img);
title('Prewitt Image');
subplot(3,3,8);
imshow(canny_img);
title('Canny Image');

```

Output:



In this image an unsharp mask is used to bring out more details by subtracting a blurry version of an image. High boost filtering allows sharpening areas but little or no sharpening in flat or smooth areas. Laplacian filter is used to highlight region of sudden intensity changes in an image like edges. Robert cross filter is used to high spatial frequency to correspond to edges. Here sobel prewitt, canny filters are used for edge detection.

Among all the filters, the unsharp mask clearly details the image. So the unsharp mask is better in the image output.